

Spectral Clustering

1. Core Formulas

a) Affinity Matrix (Similarity Graph)

Defines pairwise similarities using a Gaussian kernel (RBF):

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

σ : Bandwidth parameter controlling neighborhood size (e.g., $\sigma = 0.2$ or 1).

Alternative: Binary similarity $W_{ij} = 1$ if $\|x_i - x_j\| < \theta$, else 0 .

b) Degree Matrix

Diagonal matrix where:

$$D_{ii} = \sum_{j=1}^n W_{ij}$$

$D_{ij} = 0$ for $i \neq j$.

c) Laplacian Matrix

Normalized symmetric form (recommended):

$$L = I - D^{-1/2} W D^{-1/2}$$

I : Identity matrix.

Unnormalized form: $L = D - W$.

d) Eigen Decomposition & Clustering

1. Compute the first k eigenvectors $v_1, \dots, v_k$ of L (smallest eigenvalues).
2. Stack eigenvectors into matrix $U \in \mathbb{R}^{n \times k}$ (rows = data points in spectral space).
3. Apply **k-means** to rows of U to assign clusters (e.g., $M = 2$ or 3).

2. Numerical Example

Data

Generate 300 points from 3 Gaussian clusters:

```
import numpy as np
# Cluster 1: Mean [0, 0]
X1 = np.random.normal(0, 0.3, (100, 2))
# Cluster 2: Mean [2, 2]
X2 = np.random.normal(2, 0.3, (100, 2))
# Cluster 3: Mean [4, 0]
X3 = np.random.normal([4, 0], 0.3, (100, 2))
X = np.vstack([X1, X2, X3])
```

Steps

1. **Affinity Matrix** ($\sigma = 1$):

```

from sklearn.metrics.pairwise import rbf_kernel
W = rbf_kernel(X, gamma=1/(2*1**2)) #  $W[i,j] = \exp(-||x_i - x_j||^2 / 2)$ 
2. Degree Matrix:
D = np.diag(W.sum(axis=1))
3. Normalized Laplacian:
D_inv_sqrt = np.linalg.inv(np.sqrt(D))
L = np.eye(len(X)) - D_inv_sqrt @ W @ D_inv_sqrt
4. Eigen Decomposition ( $k = 3$ ):
eigvals, eigvecs = np.linalg.eig(L)
indices = np.argsort(eigvals)[:3] # Indices of 3 smallest eigenvalues
U = eigvecs[:, indices] # Spectral embedding (300x3 matrix)
5. k-means Clustering:
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=3).fit(U)
labels = kmeans.labels_ # Cluster assignments (0, 1, or 2)

```

Result

Original Space: Points colored by labels show perfect separation into 3 clusters.

Spectral Space: Plot of the 2nd vs. 3rd eigenvectors reveals linear separability.

3. Key Parameters & Notes

1. **Bandwidth σ :**
 - Small σ (e.g., 0.1): Captures fine-grained structure (risk: overfitting).
 - Large σ (e.g., 2): Ignores local patterns (risk: underfitting).
2. **Number of Clusters M :**
 - Set $M = k$ (eigenvectors used).
 - Estimate via eigenvalue gaps (e.g., largest jump in eigenvalues).
3. **Why Normalized Laplacian?**
 - Balances cluster sizes (prevents bias toward large clusters).
 - Outperforms unnormalized Laplacian for irregular clusters.
4. **Strengths:**
 - Handles **non-convex clusters** (e.g., sincos.txt rings).
 - Robust to noise when σ is tuned properly.

4. Handling Evidence Contradictions

Similarity Definition: uses binary thresholds (θ), while use Gaussian kernels.

→ *Resolution:* Gaussian kernels are more robust for real-world data.

Laplacian Form: uses $L = D - W$, while recommend normalized L .

→ *Resolution:* Normalized Laplacian is standard for modern implementations.

Conclusion: Spectral clustering projects data into a spectral space where clusters become separable, making it ideal for complex structures like spirals or concentric rings. Use $\sigma \approx \text{medianpairwisdistance}$ and normalized Laplacian for best results .